

Arabic Lip Reading via Spatial Strip CNN

Reframing temporal lip motion as a 2D spatial image – enabling a pure CNN to recognize spoken Arabic phrases from silent video, no audio required

Deep Learning
Exploring Unconventional Applications of CNNs

77%
Test ACCURACY

x7.7
ABOVE BASELINE

0.70
MACRO F1-SCORE

4.6M
PARAMETERS

10
ARABIC CLASSES

1 Problem Description

- Objective:** To develop a Visual Speech Recognition (Lip Reading) system for Arabic news broadcasts without relying on audio signals.
- The Challenge:** Lip motion is temporal – a sequence of frames evolving over time. Standard CNNs process spatial 2D images.
- Unconventional Approach:** Moving beyond standard static datasets (like MNIST/CIFAR) by processing dynamic video data to identify 10 highly frequent Arabic news phrases (e.g., "أَسْلَامٌ عَلَيْكُمْ") using a purely 2D-CNN architecture.

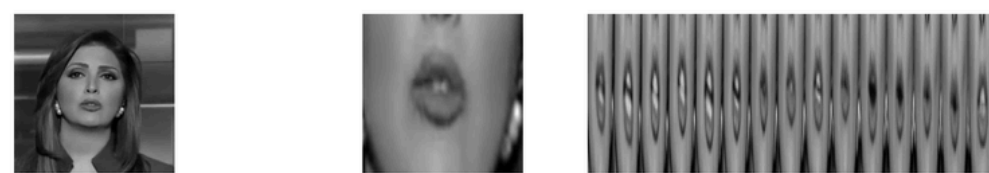
2 Why CNNs are Suitable

- Translating Time to Space:** By converting video frames over time into a single static 2D image (a "Spatial Strip"), we transformed a sequential video task into an image classification problem.
- Feature Extraction:** CNNs are exceptional at detecting local spatial patterns. In a Spatial Strip, changes in lip shapes (visemes) appear as distinct spatial textures across the image.
- Efficiency:** Standard 2D-Conv layers can effectively learn these visual speech patterns much faster and with fewer computational resources than 3D-CNNs.

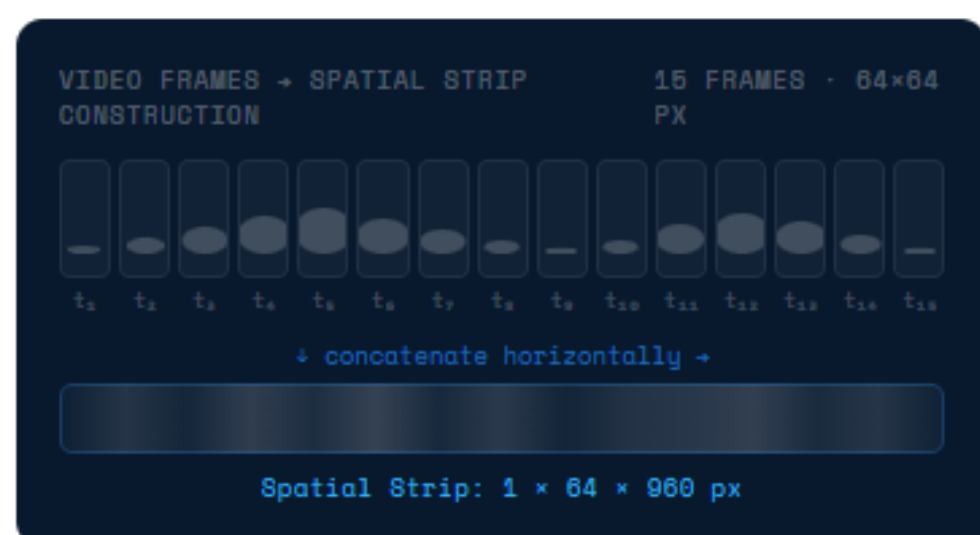
3 Data Preparation

LRC-AR Dataset Al Jazeera Arabic · Kaggle

- Face & Lip Detection:** Utilized OpenCV (Haar Cascades) to detect faces and dynamically crop the lower third to isolate the lip region.
- Spatial Strip Creation:** Sampled 15 equidistant frames from each video and concatenated them horizontally to generate a single 2D image of size **64 x 960** pixels.



- Handling Imbalance & Robustness:**
 - Applied WeightedRandomSampler to balance the 10 classes during training.
 - Implemented Data Augmentation:** Horizontal flipping, color jitter, random noise, and random erasing to prevent overfitting.



- Load Video (.mp4)
Al Jazeera news clip
- Face Detection (Haar Cascade)
Isolate face region
- Lip Crop (62%–92% of face height)
64x64 px, grayscale, CLAHE enhanced
- Concatenate 15 Frames → Strip
Shape: (1, 64, 960)
- Normalize + Augmentation
Flip · Brightness · Noise · Erasing

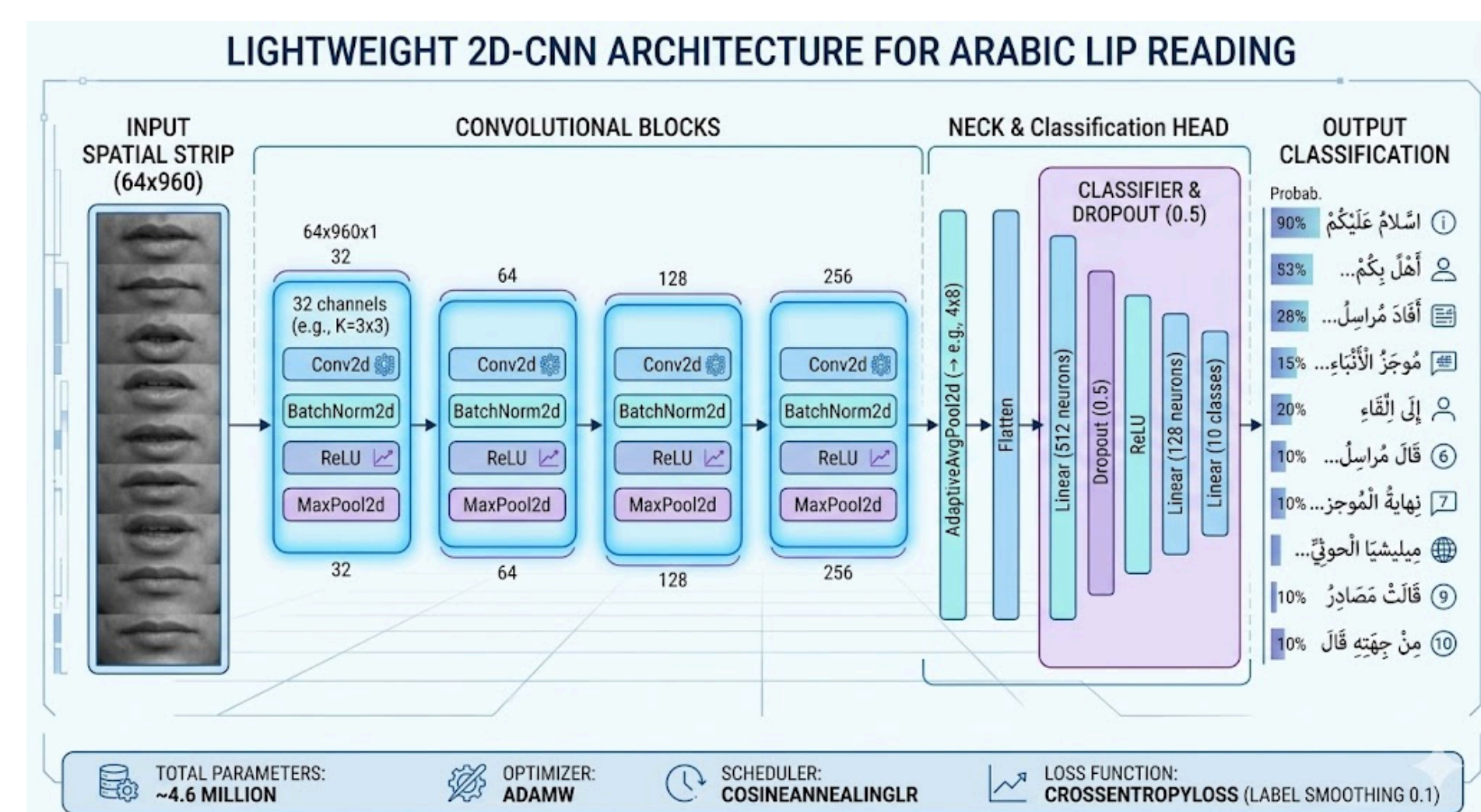
13,081
TOTAL VIDEOS

561
USED (TOP-10)

10
CLASSES

80/20
Train · Val · Test

4 Implementation Details

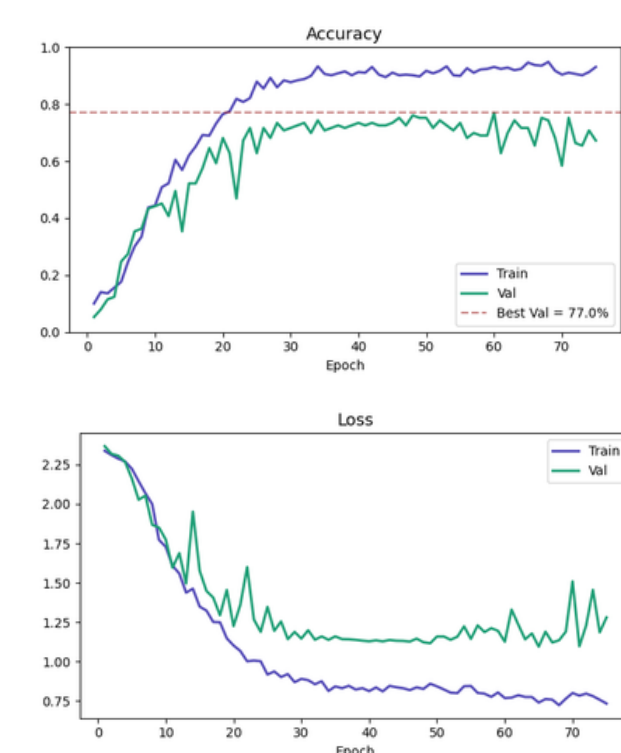


5 Results and Discussion

- Performance:**

Custom CNN vs Transfer Learning

MODEL	TEST ACC	F1	PARAMS
Custom CNN ★	77%	0.70	4.6M
ResNet18 Transfer	75%	—	11.2M



- Observations:**

- The training curves show stable convergence.
- The Confusion Matrix reveals that the model effectively distinguishes longer, visually distinct sentences (e.g., "أَسْلَامٌ عَلَيْكُمْ" achieved 86% accuracy).
- Minor misclassifications occurred mostly among sentences that share similar initial lip movements or contain fewer distinct visemes.

